Ashwin Ugale

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Skills Summary

AI / LLM: LangGraph, LangChain, RAG, Tool-Calling Agents, Hybrid Retrieval, Embeddings, OpenAI, Production LLM Agents, Multi-Tenant Tool-Calling, Eval Methodology (Precision@k, Labeled Sets), Human-in-the-Loop Workflows, Prompt Engineering
Backend & APIs: FastAPI, Node.js, Express, REST APIs, OAuth 2.0 + Proof-of-Possession, Microservices
Frontend: React, Redux, TypeScript, Vite, Tailwind CSS
Data: SQL, PostgreSQL, Redis, Qdrant, MongoDB
Infrastructure: AWS (EC2, S3, EKS, IAM), Azure, GCP, Docker, Kubernetes, Helm, Terraform, ArgoCD, GitHub Actions, CI/CD
Languages: Python, TypeScript, Java, Go, JavaScript, SQL

Experience

Enidus USA **August 2025 – Present**
Software Engineer NY, USA
*Building enterprise AI for **T-Mobile for Business** reseller operations.*

- Built a multi-tenant LLM tool-calling platform using FastAPI, OpenAI, Qdrant, and SQL Server on Azure, supporting 50+ tools with JWT-based tenant isolation and role-based access.
- Delivered 9 agentic workflows for bulk line operations, SIM/eSIM ordering, and troubleshooting, reducing 10–12 portal steps to a single chat command with human approval gates.
- Designed a hybrid retrieval layer (Qdrant semantic search + direct phone/BAN/ICCID lookups), reaching 89% intent-classification accuracy and 90.3% correct tool selection on a 442-query labeled eval.
- Migrated troubleshooting sessions from in-memory storage to a SQL-backed state machine, enabling 8 concurrent workers and preserving workflow state across deployments.
- Developed a LangGraph bill-investigation agent that analyzes billing, usage, and equipment data to explain anomalies, evaluated against a 50-query labeled dataset with approval required before corrective actions.
- Implemented OAuth 2.0 and RS256 Proof-of-Possession authentication for 20+ T-Mobile APIs, including a translation layer for legacy request formats.

Live Risk Labs **June 2025 – August 2025**
Software Engineer NY, USA

- Authored Terraform modules provisioning GCP infrastructure (VPCs, GKE clusters, IAM) for backend services.
- Built CIS-hardened base container images with Ansible; integrated into the CI pipeline for backend deploys.
- Set up GitOps deploys with ArgoCD; added Checkov for policy-as-code scanning on Terraform changes in CI.

Hyphenova **June 2024 – December 2024**
Software Engineer Intern California, USA

- Built reusable React + TypeScript components with Redux, lazy loading, and code splitting to improve frontend performance and modularity.
- Implemented Redis for caching, session handling, and rate limiting on a Node.js + MongoDB stack, reducing DB load by 40%.
- Configured GitHub Actions with Docker for testing, linting, and automated builds; enforced quality via Jest, ESLint, and Prettier with 85%+ test coverage.
- Integrated versioned container deployments to GKE with rollback-safe CI/CD pipelines.

Projects

- **DocChat** — Version-aware AI documentation assistant that retrieves answers from the exact dependency versions used by a project, preventing hallucinations caused by API version mismatches. Built on two PyPI libraries I authored: **smolAmem** (hierarchical agent memory) and **ToolPicker** (hybrid BM25 + semantic tool routing). Distributed as an MCP server (**docchat-server**) for Claude Code/Cursor and a VS Code extension. Evaluated on a labeled benchmark, achieving **94.3% version correctness** and **100% refusal on out-of-scope queries**.
- **Muteval** — "Mutation testing" for LLM evals: deliberately degrades the AI system (prompt, retrieved context, or model) and flags which regressions a team's evaluation tests fail to catch. Published to **PyPI/npm** with adapters for DeepEval, RAGAS, and promptfoo.

Certification

AWS Certified Developer – Associate | Link

Education

Indiana University Bloomington **August 2023 – May 2025**
Master of Science in Computer Science (GPA: 3.61/4) United States

K. J. Somaiya Institute of Technology **August 2019 – May 2023**
Bachelor of Technology in Electronics and Telecommunication Engineering (GPA: 8.99/10) India